

PARTS LIST

Semiconductors:

IC1	- ATmega8A MCU
IC2	- LM358 op-amp
IC3	- 7805, 5V voltage regulator
D1, D3-D7	- 1N4007 rectifier diode
D2	- 1N4148 signal diode
T1	- BC547 npn transistor
T2	- 2N2222 npn transistor
LCD1	- 16X2 LCD
LED1-LED4	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2, R7	- 1-kilo-ohm
R3	- 4.7-kilo-ohm
R4, R8,	
R12-R14, R16	- 10-kilo-ohm
R5, R6, R10, R19	- 2.2-kilo-ohm
R9, R15	- 47-kilo-ohm
R11, R17	- 3.3-kilo-ohm
R18	- 1.5-kilo-ohm
VR1	- 1-kilo-ohm potmeter
VR2, VR3	- 10-kilo-ohm potmeter

Capacitors:

C1-C4	- 0.1 μ F ceramic disk
C5	- 10 μ F, 16V electrolytic
C6	- 1000 μ F, 35V electrolytic

Miscellaneous:

X1	- 230V AC primary to 12V, 250mA secondary transformer
CT1	- 10A, 50Hz current transformer
CON1	- 6-pin connector
CON2, CON3	- 2-pin connector
S1	- Tactile switch
S2	- On/off switch
PZ1	- Piezo buzzer
RL1	- 12V, single-changeover relay
	- 28-pin IC base
	- Berg strip connectors
	- Heat-sink for 7805
	- 230V, 50Hz operated appliance

the core of CT1, which produces voltage (or EMF) proportional to the current passing through it. Voltage generated is AC in nature, which is available across points A and B. This voltage is fed to op-amp IC2, which is connected as a unity-gain amplifier, through resistor R10, potmeter VR1 and diode D2.

IC2 circuit removes the negative half of the signals received from CT1, so that only positive signals are fed to pin 27 of the MCU for analogue to digital converter (ADC). Internal ADC of the MCU measures the voltage continuously and records peak values, which are displayed on LCD1.

The circuit trips (or switches off the relay) in case current goes beyond

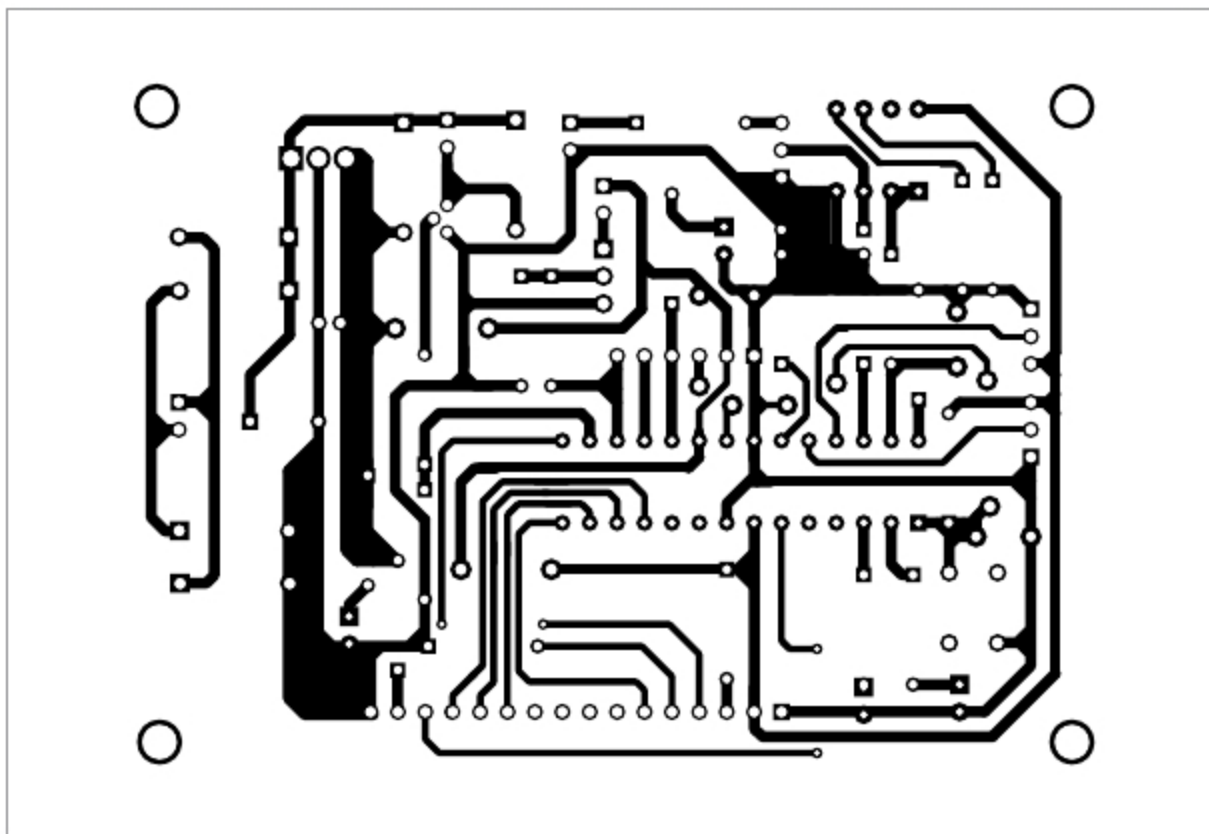


Fig. 2: Actual-size PCB layout of appliance guard

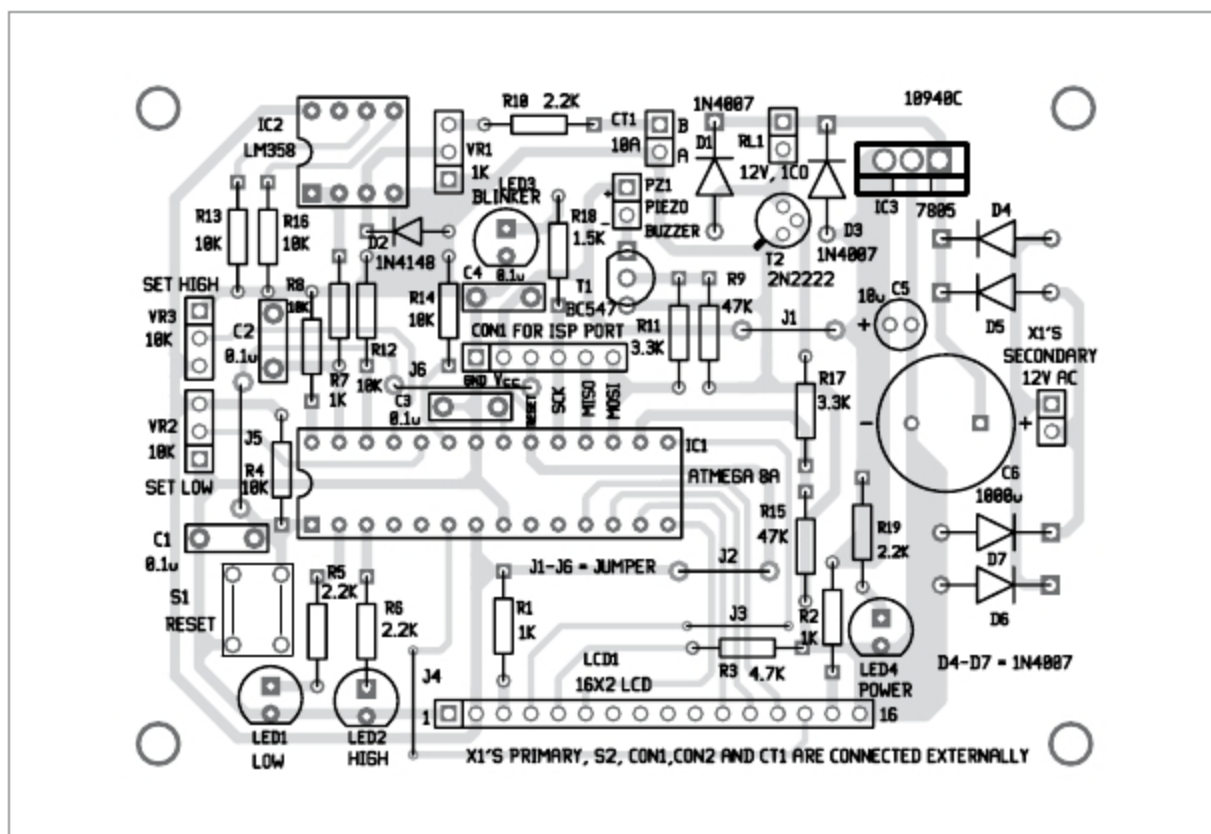


Fig. 3: Components layout for the PCB

the set high value or lower than the set low value. Tripping occurs if either high- or low-current condition continues for around 10 seconds. This protects the appliance from further damage.

Software

The software (applianceguard.c) is written in C language. AVR Studio can be used to generate the hex code (applianceguard.hex) for programming ATmega8A MCU. You can use any suitable programmer

to burn the hex code into the MCU. (At EFY Labs, ISP programmer was used for burning the hex code into ATmega8A.)

Delay can be adjusted in the code given below by modifying the value 60 to some other value.

```
#define MAX_L 60
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Construction and testing

An actual-size PCB layout of the appliance guard is shown in Fig. 2 and its components layout in Fig. 3. Assemble the circuit on the PCB and